Using Polygenic Scores in Epidemiological Practice: an Introduction

Interaction Analyses, Mediation Analyses, and a Family-Design Check

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INSTRUCTOR

Yesterday we covered missing heritability and family/twin designs. Today's session builds directly on that: once a PRS is constructed, two natural follow-up questions arise: **1)** *for whom does it matter more?* (interaction) and **2)** *through what pathway does it act?* (mediation). We will also come back to family designs at the end, because they turn out to be a powerful sanity check for both.

Learning objectives. By the end of this session you should be able to:

- Distinguish an interaction (effect modification) from a mediation (mechanism) conceptually and statistically.
- Fit, interpret, and plot an interaction model in R.
- Decompose a total effect into direct and indirect (mediated) components.
- Understand how a family (within-sibship) design can protect against confounded PRS associations.

The *cookie* scenario. We will use a simulated dataset built around one sweet story: a polygenic score for sugar addiction (`PRS_sugar`; already corrected for population stratification and scaled), weekly `exercise_hours`, the number of `cookies_per_year` eaten, and a `happiness_score` (0-100).

Two datasets. Parts 1 → 3 use `prs_cookies_happiness_individuals.csv`: 5,000 unrelated individuals, no family structure. Part 4 switches to `prs_cookies_happiness_families.csv`: 1,500 families of 2 siblings (3,000 individuals), which adds an unmeasured family-level confounder so we can demonstrate the potential of family designs to correct for confounding effects.

1 Warm-up: exploring the data

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Before touching any model, let's just look at the data; always the first reflex with a new dataset. Note that for this part and the next two (interaction, mediation), we work with the **individuals** dataset.

Q1.1. Load `prs_cookies_happiness_individuals.csv` into R and produce summary statistics (mean, SD, range) for `PRS_sugar`, `exercise_hours`, `cookies_per_year`, and `happiness_score`. Do the ranges look plausible?

Q1.2. Using `ggplot2`, plot a histogram of `PRS_sugar` and a scatterplot of `cookies_per_year` against `PRS_sugar`. Does a simple linear association seem present?

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2 Interaction analyses

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Concept. An interaction (effect modification) asks: does the association between the PRS and an outcome change across levels of another variable? Here: does exercise buffer the effect of the sugar-addiction PRS on cookie consumption? We continue with the **individuals** dataset.

Q2.1. Fit a linear model predicting `cookies_per_year` from `PRS_sugar`, `exercise_hours`, and their product term. Write the model equation you are fitting.

Q2.2. Interpret the coefficient on the interaction term. Is it statistically significant? In plain language, what does its sign tell you about how exercise modifies the PRS effect?

Q2.3. Compute the predicted PRS-to-cookies slope for three exercise levels: low (mean − 1 SD), average (mean), and high (mean + 1 SD). I.e., how much `cookies_per_year` changes per unit increase in `PRS_sugar`. This is the "simple slopes" approach.

HINT

```r
m_int <- lm(cookies_per_year ~ PRS_sugar * exercise_hours, data = data)
summary(m_int)

ex_levels <- c(mean(data$exercise_hours) - sd(data$exercise_hours),
               mean(data$exercise_hours),
               mean(data$exercise_hours) + sd(data$exercise_hours))

b <- coef(m_int)
slopes <- b["PRS_sugar"] + b["PRS_sugar:exercise_hours"] * ex_levels
data.frame(exercise = ex_levels, slope = slopes)
```

Q2.4. Now plot the interaction: predicted `cookies_per_year` against `PRS_sugar`, with three separate lines/colors for low, average, and high exercise. Does the picture match your simple-slopes numbers?

HINT

```r
data$ex_group <- cut(data$exercise_hours,
                     breaks = quantile(data$exercise_hours, c(0, 1/3, 2/3,
                         1)),
                     labels = c("Low", "Average", "High"),
                     include.lowest = TRUE)

ggplot(data, aes(x = PRS_sugar, y = cookies_per_year, color = ex_group)) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(color = "Exercise level")
```

Q2.5. *Conceptual, no code.* Someone tests the PRS × exercise interaction on "high vs. low cookie consumption" using logistic regression and finds no significant interaction on the odds-ratio scale, even though your model above showed a clear one. Is this a contradiction? Why or why not?

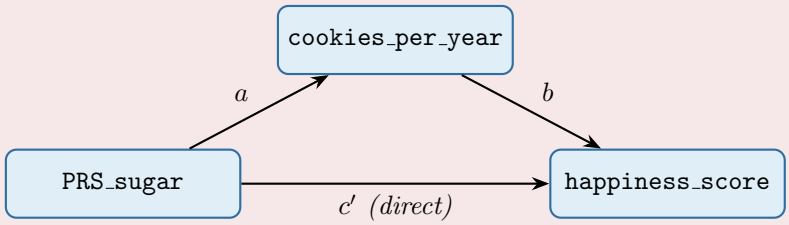
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3 Mediation analyses

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Concept. Mediation asks a different question: not "for whom is the effect bigger", but "through what pathway does the effect operate?". Here: does the PRS affect happiness *because* it drives cookie consumption, or does it also have some effect on happiness that bypasses cookies entirely? Still the **individuals** dataset here.

The DAG. Three variables, three arrows:



`PRS_sugar` (X) is the exposure, `cookies_per_year` (M) is the mediator, `happiness_score` (Y) is the outcome. Three path coefficients matter:

- a: the effect of X on the mediator M (path a, Q3.2 below).
- b: the effect of M on Y, holding X fixed (path b, part of Q3.3).
- c': the *direct* effect of X on Y, holding M fixed; the part of the effect that does *not* run through cookies (also Q3.3).

From these we get the *indirect / mediated* effect, $a \times b$: how much of X's effect on Y is transmitted through M. And critically:

$$\underbrace{c}_{\text{total effect}} = \underbrace{c'}_{\text{direct effect}} + \underbrace{a \times b}_{\text{indirect effect}}$$

The total effect c is what you get from a simple regression of Y on X alone (Q3.1, no M in the model), ignoring the pathway entirely. Decomposing it into c' and $a \times b$ is the whole point of a mediation analysis: it tells you not just *whether* the PRS matters for happiness, but *how much of that effect is explained by cookie consumption specifically.*

Q3.1. Fit the "total effect" model: `happiness_score` ~ `PRS_sugar`. Report and interpret the coefficient.

Q3.2. Fit *path a*: `cookies_per_year` ~ `PRS_sugar`. Report the coefficient.

Q3.3. Fit *path b / direct effect*: `happiness_score` ~ `cookies_per_year` + `PRS_sugar`. Report both coefficients.

HINT

```r
m_total <- lm(happiness_score ~ PRS_sugar, data = data)
m_a     <- lm(cookies_per_year ~ PRS_sugar, data = data)
m_bcp   <- lm(happiness_score ~ cookies_per_year + PRS_sugar, data = data)

coef(m_total)["PRS_sugar"]    # total effect (c)
coef(m_a)["PRS_sugar"]        # path a
coef(m_bcp)["cookies_per_year"]  # path b
coef(m_bcp)["PRS_sugar"]      # direct effect (c')
```

Q3.4. Compute the indirect (mediated) effect as $a \times b$. Compare it to the total effect from Question 1. What proportion of the total effect is mediated through cookies?

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The four assumptions. Everything above (path a, path b, and the decomposition $c = c' + a \times b$) only has a **causal** interpretation if a few assumptions hold:

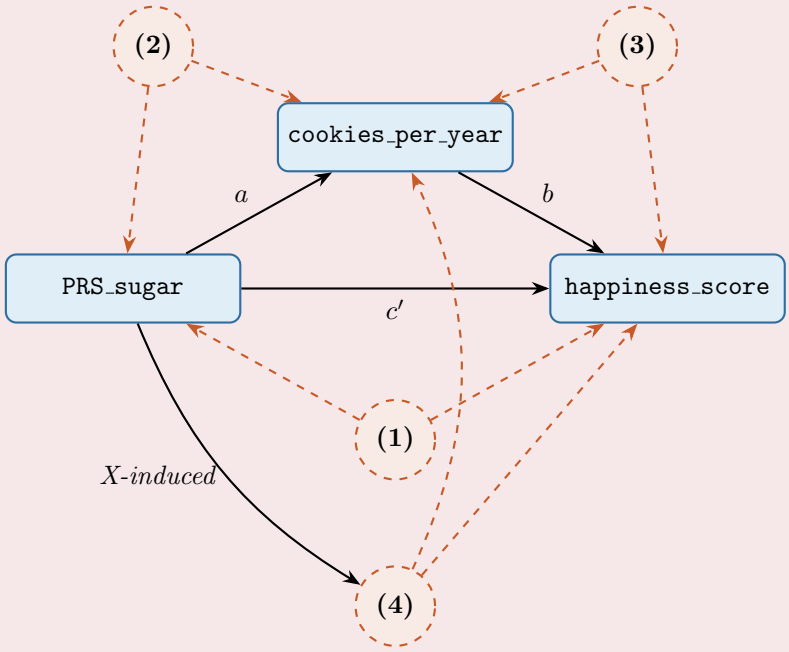
(1) No unmeasured $X \to Y$ confounding. Nothing besides X itself should be driving both `PRS_sugar` and `happiness_score` directly.

(2) No unmeasured $X \to M$ confounding. Nothing should be driving both `PRS_sugar` and `cookies_per_year` other than the causal arrow between them.

(3) No unmeasured $M \to Y$ confounding. Nothing besides M itself should be driving both `cookies_per_year` and `happiness_score`. Anything correlated with eating habits and wellbeing (general health consciousness, stress, socioeconomic status, family environment) can potentially confound path b even when X is as clean as a PRS.

(4) No X-induced confounder of $M \to Y$. There should be no variable that (a) is itself affected by `PRS_sugar`, and (b) also confounds the `cookies_per_year` → `happiness_score` relationship. For example: if the PRS shapes early-life dietary habits, and those early habits *also* independently affect adult happiness through some other channel, simple product-of-coefficients mediation breaks down.

All four, on one picture. Solid arrows are the causal paths we actually want to estimate (plus the side-effect of the PRS in assumption (4)). Dashed arrows are the confounding paths each assumption requires to be *absent*:



Nodes (1), (2) and (3) are generic unmeasured confounders sitting on each edge of the triangle; node (4) is different in kind, since X *genuinely* causes it (solid arrow). If every dashed arrow above is truly absent, the decomposition $c = c' + a \times b$ has a clean causal reading.

HINT

For a proper analysis framework with confidence intervals and (if needed) exposure-mediator interaction and survival outcomes, packages such as `mediation` can be very helpful:

```
# install.packages("mediation")
library(mediation)
med_fit <- lm(cookies_per_year ~ PRS_sugar, data = data)
out_fit <- lm(happiness_score ~ cookies_per_year + PRS_sugar, data = data)
med_out <- mediate(med_fit, out_fit, treat = "PRS_sugar",
                   mediator = "cookies_per_year", boot = TRUE, sims = 500)
summary(med_out)
```

Q3.5. When using the mediation package, do you recover similar results as done previously?

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4 A family-design robustness check

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This is where we tie back to what you found in Part 3. Do you remember assumption (3) from the mediation talk (→ no unmeasured $M \to Y$ confounding)? We'll deliberately violate assumption (3) and see what it does to `cookies_per_year` → `happiness_score` (path b), and then use a family design to fix it.

We switch datasets here. Load `prs_cookies_happiness_families.csv` instead: 1,500

families of 2 siblings. I've built in a family-level factor called `family_treat_culture`: household norms around treats and snacking, shared by siblings, which affects *both* `cookies_per_year` *and* `happiness_score` directly; exactly an assumption-(3) violation. As part of this exercise, I am *giving you* this confounder as an actual column, so you can check with your own hands whether a within-family design really does correct for it. In practice **you almost never get to observe such a confounder directly**.

A separate warning, before you even get to confounding. The naive model below (`happiness_score` ∼ `cookies_per_year` + `PRS_sugar`) treats all 3,000 rows as independent observations, but siblings share unmeasured family-level factors, so their residuals are correlated. That makes ordinary `lm()` standard errors too small, and your p-values and confidence intervals overconfident, *regardless of whether any confounding is present at all.* This is separate from confounding: confounding biases the *estimate itself*, while clustering biases the *uncertainty around it.* Family fixed effects happen to fix both here; otherwise, use cluster-robust standard errors (e.g. `sandwich` or `clubSandwich`) rather than the default ones.

Q4.1. Load `prs_cookies_happiness_families.csv`. Fit the naive model: `happiness_score` ∼ `cookies_per_year` + `PRS_sugar` (the same specification as Part 3, Q3.3). Report the `cookies_per_year` coefficient.

Q4.2. Now fit a within-family model by adding family fixed effects: `happiness_score` ∼ `cookies_per_year` + `PRS_sugar` + `factor(family_id)`. Report the `cookies_per_year` coefficient.

HINT

```r
data_fam <- read.csv("prs_cookies_happiness_families.csv")

m_naive  <- lm(happiness_score ~ cookies_per_year + PRS_sugar, data = data_
    fam)
m_within <- lm(happiness_score ~ cookies_per_year + PRS_sugar + factor(
    family_id), data = data_fam)

coef(m_naive)["cookies_per_year"]
coef(m_within)["cookies_per_year"]

# Optional: at least fix the standard errors of the naive model, even
    without
# correcting the point estimate, using cluster-robust SEs:
# install.packages("sandwich"); install.packages("lmtest")
# library(sandwich); library(lmtest)
# coeftest(m_naive, vcov = vcovCL(m_naive, cluster = data_fam$family_id))

# Alternative to family fixed effects, sibling-difference approach:
# library(dplyr)
# wide <- data_fam %>% group_by(family_id) %>%
#   summarise(d_cook = diff(cookies_per_year), d_prs = diff(PRS_sugar),
#             d_happy = diff(happiness_score))
# lm(d_happy ~ d_cook + d_prs, data = wide)
```

Q4.3. Compare the two coefficients (including their *sign*), and also compare the within-family estimate to path b from Part 3, Q3.3 (on the unrelated individuals dataset). What does this tell you about the presence (or absence) of family-level confounding, and about whether the family design recovered the "true" effect?

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Now the interesting part. You have a column called `family_treat_culture` in this dataset (household norms around treats/snacking, shared within each family). Suppose we treat it as if it were a genuinely measured confounder (in most real studies you would never be so lucky). Try adjusting for it directly in an ordinary regression, and compare that estimate to your within-family estimate above.

Q4.4. Fit `happiness_score` ~ `cookies_per_year` + `PRS_sugar` + `family_treat_culture`. Report the `cookies_per_year` coefficient, and compare it to (a) the naive estimate, and (b) the within-family estimate from Q4.2.

Q4.5. Given what you just observed, why does that make the within-family design valuable when you can't measure or even name the confounder?

Q4.6. *Bonus, if time allows.* Repeat the sibling-difference regression from the Q4.2 hint above. Does it agree with the family-fixed-effects and direct-adjustment estimates?

Q4.7. *Synthesis.* Given what you just found, would you trust the mediation decomposition from Part 3 as-is if it had been run on this families dataset instead? What would re-running it with some form of family adjustment require in terms of data?

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5 Wrap-up discussion

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Let's close with a few last open questions. No code needed here.

Q5.1. In your own research or educational area, name a variable you would treat as a moderator versus one you would treat as a mediator of a PRS effect. What made you classify it that way?

Q5.2. In practice, interaction tests typically require much larger sample sizes than main-effect tests. What does this imply for interpreting a "null" PRS × environment interaction in a modestly-sized study?

Q5.3. Be honest: how does incorporating a family design sound for your own research or studies? Would you feel confident using family designs in the future?

End of workshop. Thank you!